

Complete Constraint Reference (Full Detail)

Level 1: Sequence Validity

SequenceValidity

Weight: 1.0 | **Mode:** HARD_GATES | **Threshold:** < 0.1

What it detects: Non-standard amino acid characters (e.g., X, *, numbers, spaces) in either WT or mutant sequence

What it means: Ensures both sequences contain only valid IUPAC 20-letter amino acid alphabet. Invalid characters would corrupt all downstream physicochemical calculations.

Score range: 0 = all valid, 1 = any invalid

Fallback strategy: None (fatal if failed)

Source: IUPAC-IUB Joint Commission on Biochemical Nomenclature

LengthMatch

Weight: 1.0 | **Mode:** HARD_GATES | **Threshold:** < 0.1

What it detects: Length difference between WT and mutant (insertions or deletions)

What it means: All constraints assume residue-by-residue alignment. Indels break this assumption and require separate handling. This enforces point-mutation-only analysis.

Score range: 0 = same length, 1 = any difference

Fallback strategy: None (fatal if failed)

Source: Framework design decision

NoStopCodons

Weight: 1.0 | **Mode:** HARD_GATES | **Threshold:** < 0.1

What it detects: Stop codon characters (X, *) in the mutant sequence

What it means: Stop codons represent truncated or non-functional proteins. Their presence in a designed mutant is almost always undesirable for functional protein engineering.

Score range: 0 = none, 1 = all positions are stops

Fallback strategy: None

Source: Standard genetic code

Level 2: Physicochemical Compatibility

GranthamDistance

Weight: 0.3 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Chemical dissimilarity between WT and mutant amino acids

What it means: Grantham combines composition (C), polarity (P), and volume (V) into a single distance. Large Grantham distances (e.g., Cys->Trp = 215) indicate chemically radical substitutions likely to disrupt local structure.

Score range: 0 = identical, 1 = maximum distance (215)

Fallback strategy: None (pure sequence-based)

Source: Grantham R. (1974) Science 185:862-864

BLOSUM62Score

Weight: 0.3 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Substitution acceptability based on observed evolutionary frequencies

What it means: BLOSUM62 scores reflect how often a substitution is observed in conserved protein blocks. Positive scores (e.g., Leu->Ile = +2) = common/acceptable. Negative scores (e.g., Cys->Trp = -3) = rare/deleterious. This captures evolutionary 'wisdom' about viable substitutions.

Score range: 0 = highly conservative (BLOSUM +11), 1 = highly radical (BLOSUM -4)

Fallback strategy: None (pure sequence-based)

Source: Henikoff S. & Henikoff J.G. (1992) PNAS 89:10915-10919

HydropathyChange

Weight: 0.2 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Changes in hydrophobic/hydrophilic character at mutated positions

What it means: Hydropathy drives protein folding (hydrophobic effect). Flipping a buried hydrophobic residue to charged (e.g., Ile->Arg) can destabilize the core. Preserving hydropathy maintains the thermodynamic driving force for native structure.

Score range: 0 = no change, 1 = maximum flip (e.g., Arg->Ile)

Fallback strategy: None (pure sequence-based)

Source: Kyte J. & Doolittle R.F. (1982) J. Mol. Biol. 157:105-132

SecondaryStructurePropensity

Weight: 0.2 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Changes in intrinsic alpha-helix, beta-sheet, or turn propensity

What it means: Each amino acid has intrinsic tendencies to form particular secondary structures. Mutating a strong helix-former (Glu, P_alpha=1.53) to a strong breaker (Pro, P_alpha=0.59) can destroy a helix. This constraint preserves the propensity landscape.

Score range: 0 = same propensity class, 1 = maximal shift in [P_alpha, P_beta, P_turn] space

Fallback strategy: None (pure sequence-based)

Source: Chou P.Y. & Fasman G.D. (1974) Biochemistry 13:222-245

AggregationPropensity

Weight: 0.3 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Increase in intrinsic aggregation tendency of mutated residues

What it means: Some amino acids (Phe, Ile, Leu, Val, Trp, Tyr) are strongly aggregation-prone. Introducing them can trigger amyloid-like or amorphous aggregation. This constraint only penalizes INCREASES in aggregation propensity (asymmetric).

Score range: 0 = no increase, 1 = maximal increase

Fallback strategy: None (pure sequence-based)

Source: Fernandez-Escamilla et al. (2004) Nature Biotech 22:1302; Pawar et al. (2005) JMB 350:379

BurialChange

Weight: 0.2 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Mismatch between expected solvent accessibility of WT vs mutant residue

What it means: Buried positions need small/hydrophobic residues; surface positions tolerate larger/charged ones. If SASA data is available, this checks RSA compatibility. Otherwise falls back to hydropathy as a proxy for burial preference.

Score range: 0 = compatible burial, 1 = severe mismatch

Fallback strategy: Hydropathy difference proxy

Source: Tien M.Z. et al. (2013) PLoS ONE 8(2):e80635 (MAX_ASA values)

ESM2_wt_marginal (optional)

Weight: 1.0 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Plausibility of mutant sequence under ESM-2 protein language model

What it means: ESM-2 was trained on millions of natural protein sequences and learns statistical patterns of viable sequences. A large drop in pseudo-log-likelihood (PLL) suggests the mutant is 'unnatural' and likely non-functional. This is a learned, data-driven constraint.

Score range: 0 = mutant as likely as WT, 1 = mutant highly improbable

Fallback strategy: BLOSUM62-based proxy if ESM-2 unavailable

Source: Lin Z. et al. (2023) Science 379:1123-1130

Level 3: Structural Integrity

FoldIntegrity

Weight: 1.0 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Loss of structural fold confidence upon mutation

What it means: If AlphaFold pLDDT is provided: scores drop below 70 (unreliable structure). If not: uses sequence complexity (Shannon entropy) as proxy — low-complexity sequences (e.g., poly-Ala) tend to be unstructured.

Score range: 0 = confident fold, 1 = unfolded/low complexity

Fallback strategy: Sequence complexity (Shannon entropy)

Source: Jumper J. et al. (2021) Nature 596:583-589 (AlphaFold)

AllostericNetworkConservation

Weight: 0.8 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Disruption of allosteric communication pathways or network hubs

What it means: Allosteric sites are distant from active sites but essential for regulation. Mutating them can abolish cooperative behavior. If annotated sites unavailable, uses structural proxies: contact density, conservation scores, secondary structure state, and hydrophobicity.

Score range: 0 = no disruption, 1 = severe hub disruption

Fallback strategy: Contact density + conservation + SS state + hydrophobicity proxy

Source: Ribeiro A.A.S.T. & Ortiz V. (2015) PNAS 112:6056-6061

ConformationalDynamics

Weight: 0.6 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Altered flexibility profile at mutated positions

What it means: Protein function often depends on dynamic motions (hinges, loops). Glycine is highly flexible; Proline is rigid. Mutating between them can freeze or over-flexibilize a region. If MD RMSF data available: compares flexibility changes directly.

Score range: 0 = same flexibility, 1 = extreme change (Gly->Pro or vice versa)

Fallback strategy: Glycine/Proline flexibility proxy

Source: General MD/RMSF literature (McCammon et al. 1977)

SecondaryStructureIntegrity

Weight: 0.5 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Secondary structure state transitions (helix<->sheet<->coil)

What it means: DSSP-assigned secondary structure reflects actual 3D geometry. A helix-to-sheet transition at a mutation site can destroy local packing. If DSSP unavailable, uses Chou-Fasman propensity class changes as proxy.

Score range: 0 = no transitions, 1 = all mutations cause transitions

Fallback strategy: Chou-Fasman propensity class changes

Source: Kabsch W. & Sander C. (1983) Biopolymers 22:2577-2637 (DSSP)

Level 4: Evolutionary Conservation

EvolutionaryCoupling

Weight: 1.0 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Mutations at positions that are evolutionarily co-varying

What it means: Direct Coupling Analysis (DCA) identifies residue pairs that co-evolve — they likely interact physically or functionally. Mutating BOTH members of a coupled pair is especially damaging. If EC scores unavailable, uses BLOSUM penalty amplified by contact density.

Score range: 0 = no coupled mutations, 1 = strongly coupled pair both mutated

Fallback strategy: BLOSUM penalty x contact density x conservation

Source: Marks D.S. et al. (2011) PLoS ONE 6:e28766 (GREMLIN/EVcouplings)

PhylogeneticConservation

Weight: 0.8 | **Mode:** HYBRID | **Threshold:** < 0.5

What it detects: Mutations at phylogenetically conserved positions

What it means: Positions conserved across orthologs are under strong purifying selection — they are critical for structure or function. If ConSurf/Rate4Site scores available: high conservation = high penalty. Otherwise: BLOSUM serves as a proxy (conserved positions have few acceptable substitutions).

Score range: 0 = variable position, 1 = highly conserved position mutated

Fallback strategy: BLOSUM-based conservation proxy

Source: Pupko T. et al. (2002) Bioinformatics 18:S71 (Rate4Site/ConSurf)

Level 5: Functional / Phenotypic

CatalyticSiteIntegrity

Weight: 1.0 | **Mode:** HARD_GATES | **Threshold:** < 0.3

What it detects: Mutations at known catalytic or active site residues

What it means: Catalytic residues (His, Asp, Glu, Cys, Lys, Ser, Tyr) perform chemistry. Mutating them almost always abolishes activity. If CSA/UniProt annotations available: any hit is penalized severely. Otherwise: penalizes mutations of common catalytic residue types.

Score range: 0 = no catalytic hit, 1 = all mutations at catalytic sites

Fallback strategy: Common catalytic residue type penalty (H,D,E,C,K,S,Y)

Source: Catalytic Site Atlas (CSA) / UniProt functional annotations

BindingInterfaceDisruption

Weight: 0.9 | **Mode:** HARD_GATES | **Threshold:** < 0.3

What it detects: Mutations at protein-protein binding interface residues

What it means: Interface residues mediate molecular recognition. Mutating them can abolish binding to partners. Hydrophobic-to-polar mutations at interfaces are particularly damaging because they disrupt the hydrophobic core of the interface. If interface annotations available: direct hit penalty.

Score range: 0 = no interface disruption, 1 = all mutations at interface

Fallback strategy: Hydrophobic-to-polar penalty at surface

Source: General protein interface analysis literature